



# The Development of Collision-Induced Dissociation Gas Cell (CID-GC) for Quantum Sensing of Nuclei



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## Challenges to Beam Purity and Rate in Low-Energy Physics

- Short-lived molecular ions require radioactive beam facilities
- Beam facilities do not typically provide the molecules of interest, while **molecular contaminants** pose an increasing challenge to stopped beam experiments
- Molecular formation with stopped rare isotopes leads to signal splitting, reducing the effective signal in target mass channels [1]
- **Space-charge effects**: high-intensity beams generate significant He<sup>+</sup>/e<sup>-</sup> pairs, creating space-charge fields that hinder ion transport and reduce extraction efficiency [2]

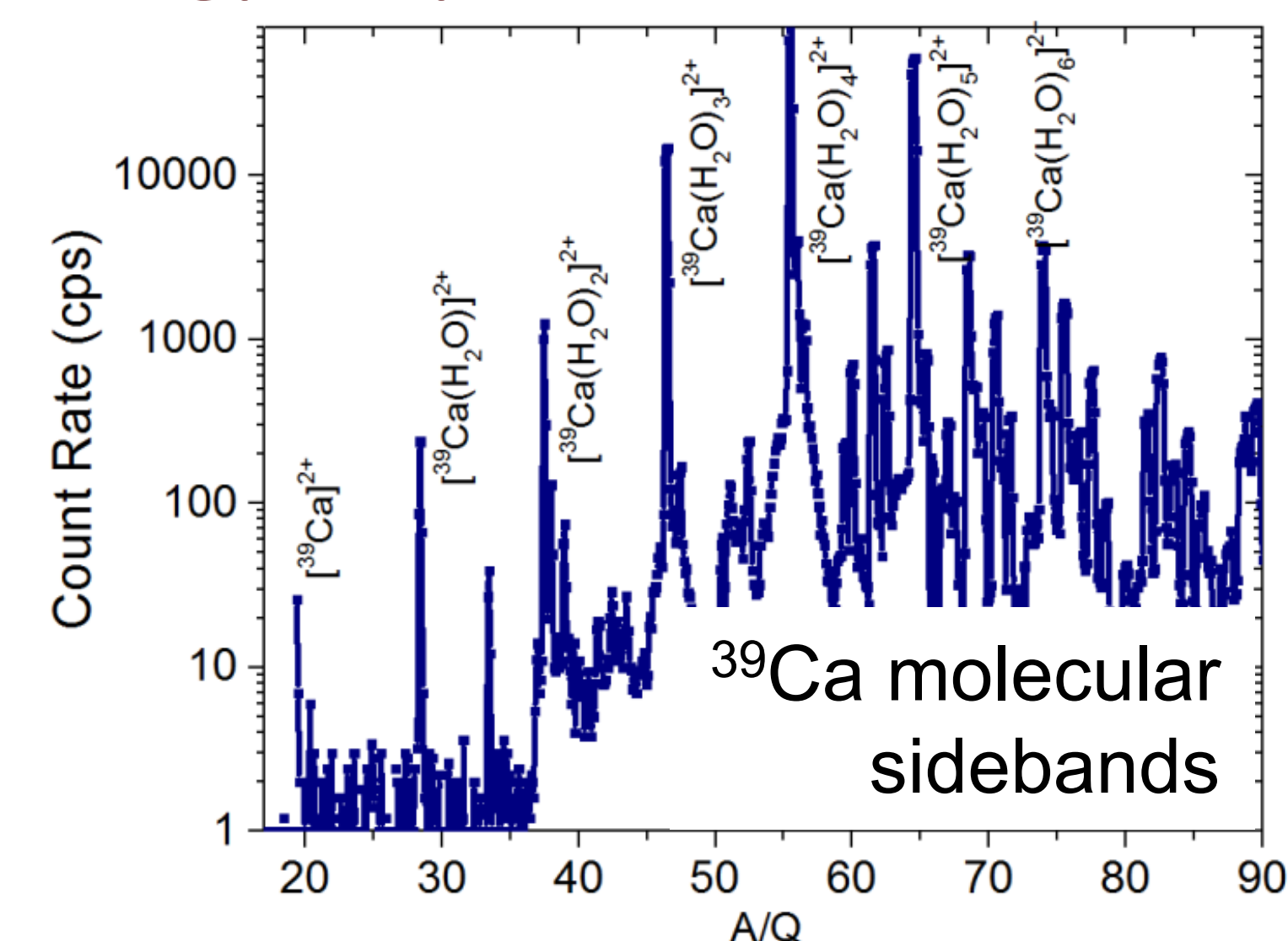


Fig 1. Example of a molecular contaminant spectrum (Courtesy to R. Ringle).

## Collision-Induced Dissociation Gas Cell (CID-GC)

- Various groups (e.g. at GSI [3] and TRIUMF [4]) have applied CID in RFQ cooler/bunchers with varying degrees of success and efficiency, while laser dissociation is molecule specific and requires extensive knowledge of the molecule
- With the LEBIT group at FRIB, we are developing a general-purpose device to **break up molecules** through collision-induced dissociation [1]
- Molecules pass through an ultra-thin Si<sub>3</sub>N<sub>4</sub> membrane to induce dissociation, followed by RFQ for trapping and/or molecule formation
- Molecular spectroscopy performed at Texas A&M in new general-purpose action spectrometer

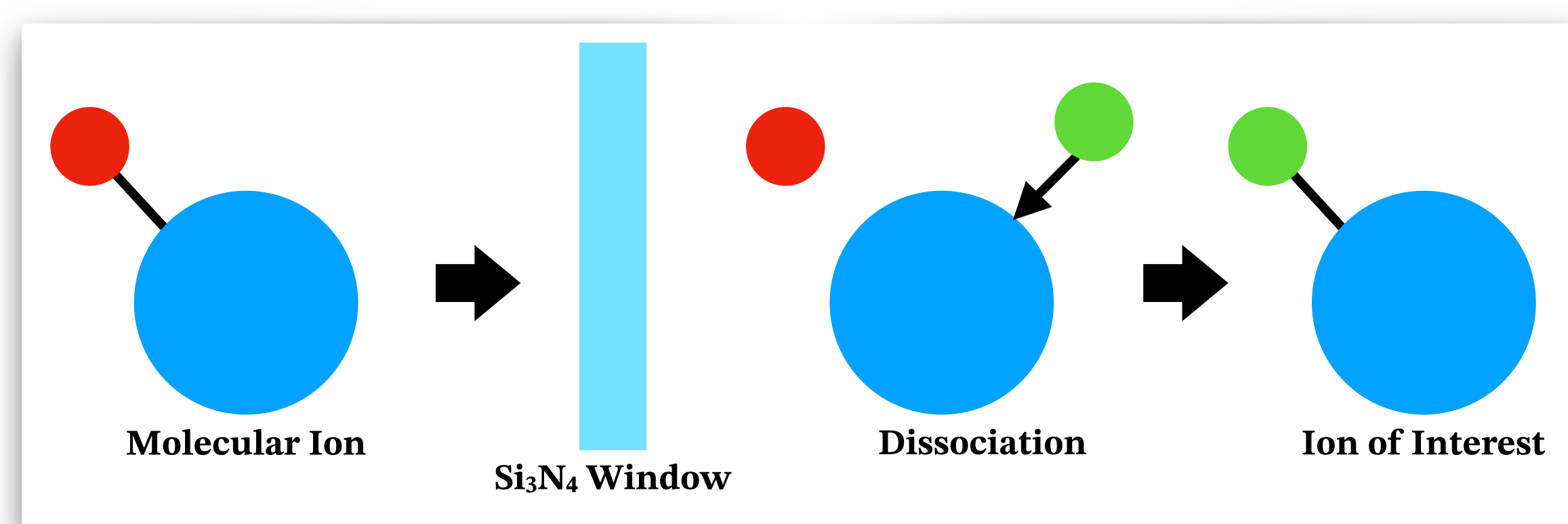


Fig 2. Illustration of the dissociation + formation mechanism.

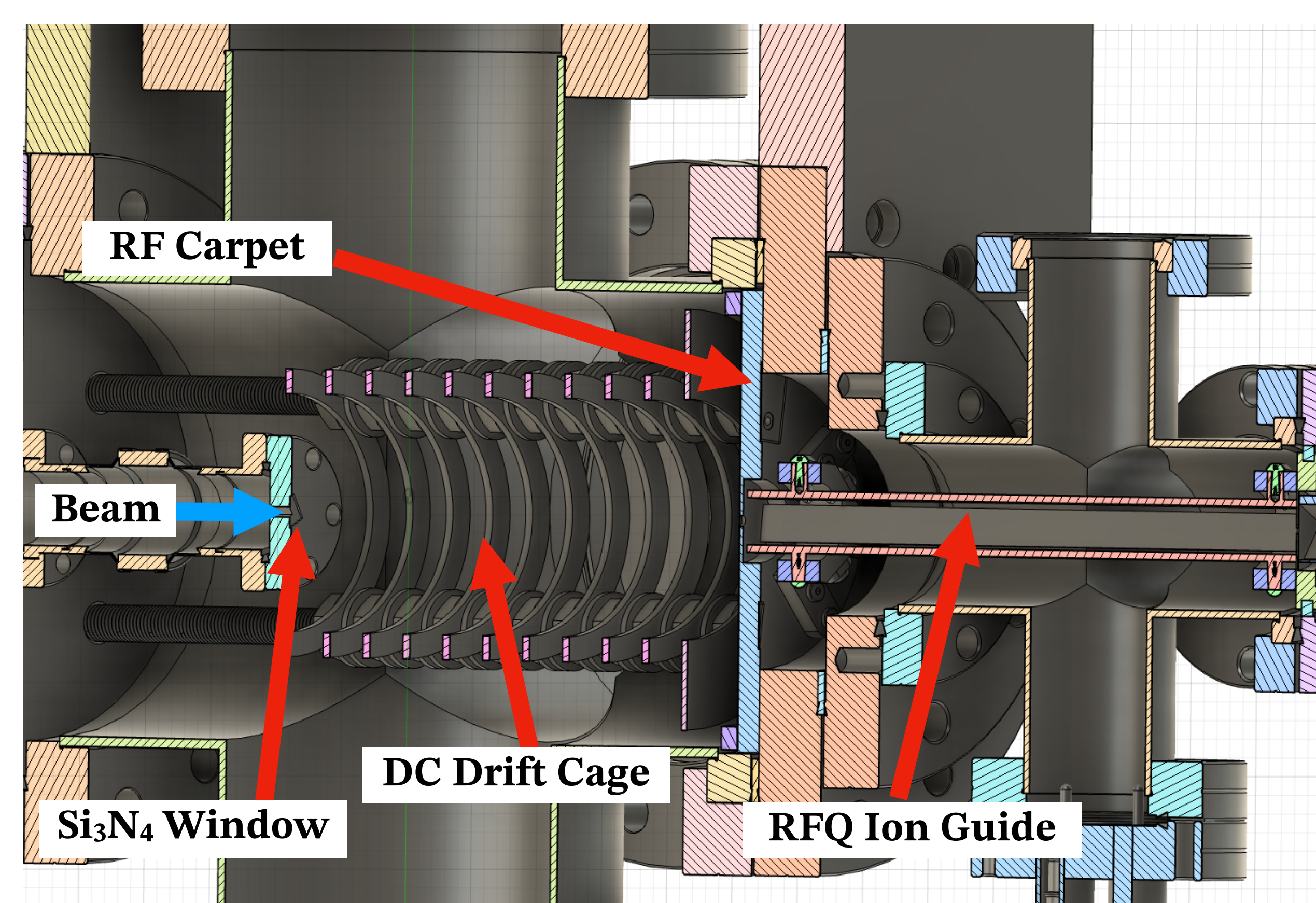


Fig 3. Cross-section of the CID-GC [1].

## The Si<sub>3</sub>N<sub>4</sub> Dissociation Window

- **2 mm × 2 mm** Si<sub>3</sub>N<sub>4</sub> window, just **~21 nm thick** (≈ 70 atomic layers)
- Impinging molecular beam get ro-vibrationally excited through collisions far above the molecular **dissociation** threshold
- Withstands pressure differentials up to 200 mbar → allows for gas-filled **RFQ trapping section** after dissociation for efficient cooling of **isotopes of interest** OR the creation of **molecules of interest**

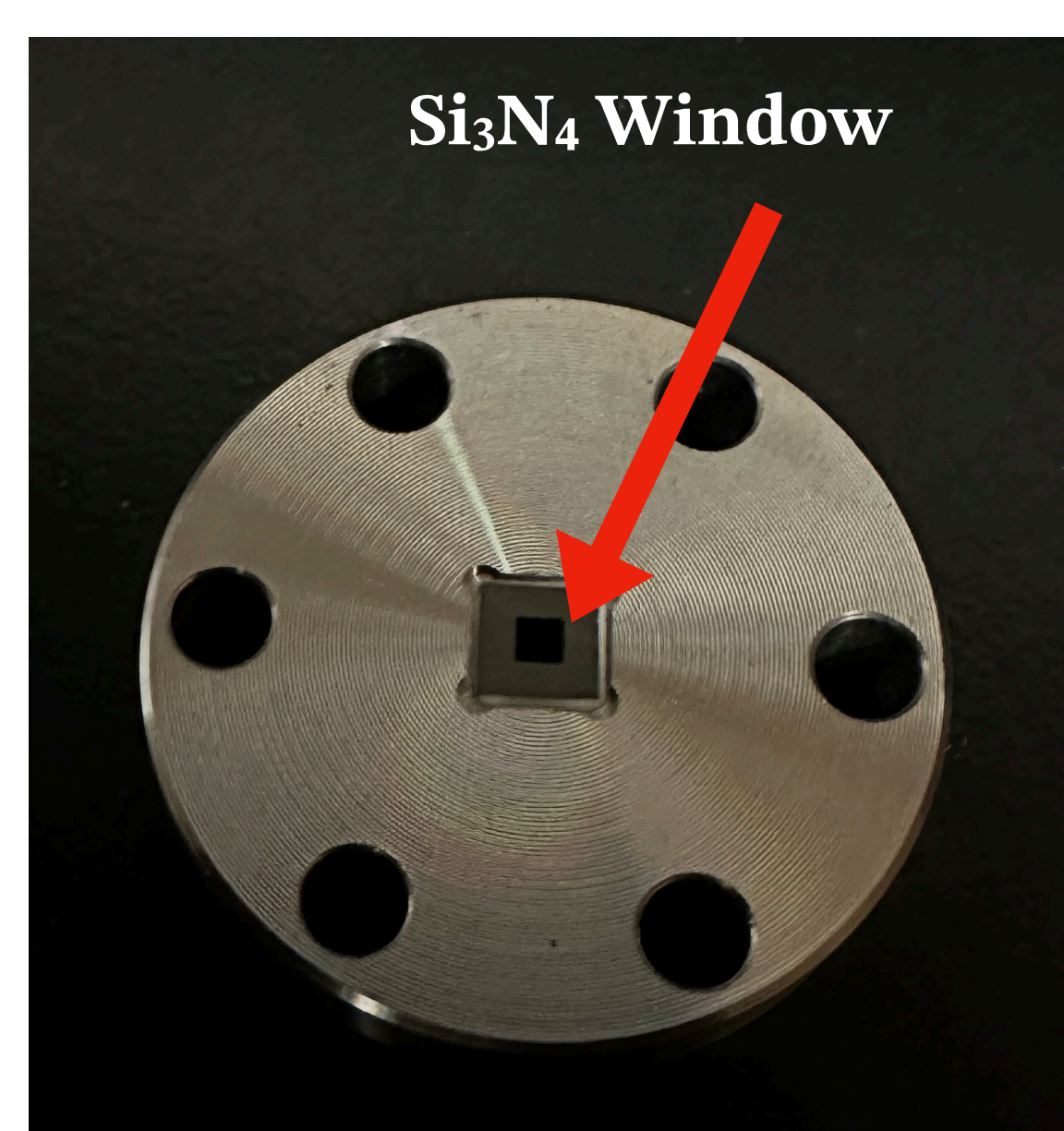


Fig 4. The Si<sub>3</sub>N<sub>4</sub> window installed on a CF flange [1].

## The RF Carpet

- After dissociation, ions are drifted by DC fields and guided across a **4-phase RF carpet** using "ion surfing" (MHz-frequency fields)
- Guides ions from the CID-GC into RFQ ion guides for further purification and detection. [1]

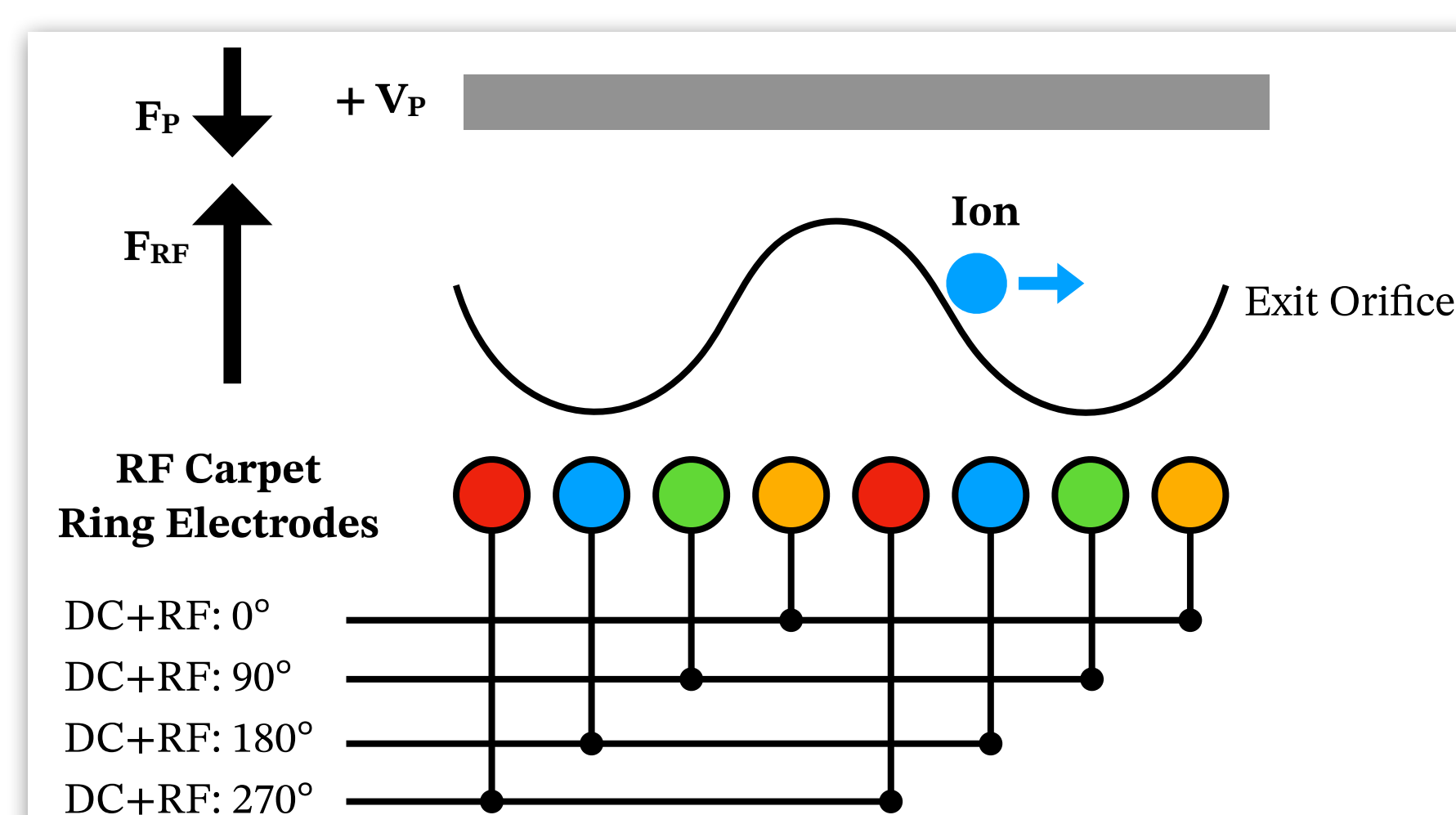


Fig 5. Illustration of how 4-phase ion surfing works.

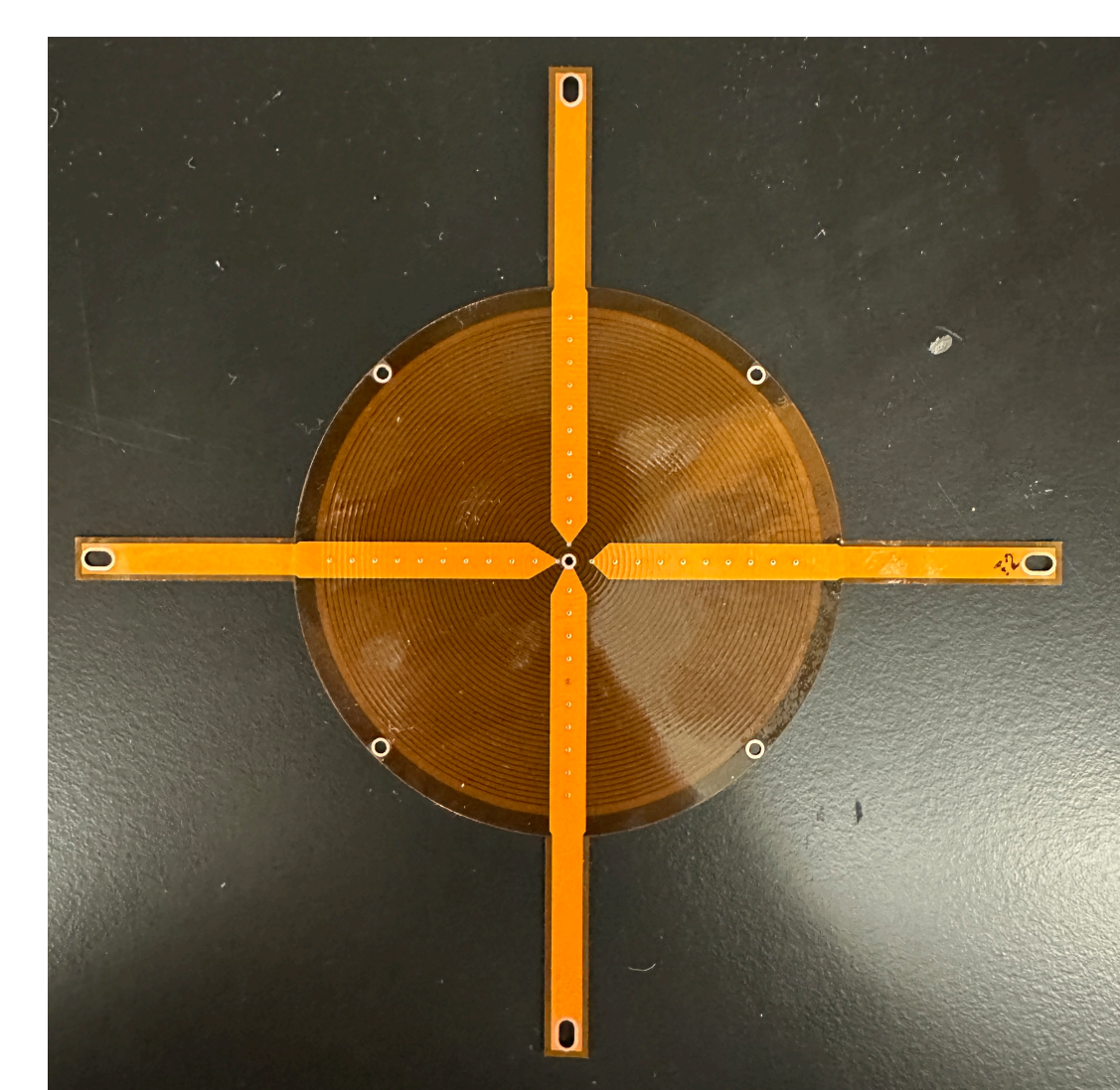


Fig 6. The RF carpet.

## Quantum Sensing of Radioactive Molecules at NEPTUNE

- The Standard Model fails to describe 95% of the mass and energy in the Universe, matter-antimatter imbalance, ...
- Our group is searching for violations of fundamental symmetries in radioactive molecules [1]
- Trapping (radioactive) molecular ions at **NEPTUNE** in a Penning ion trap can amplify electroweak nuclear properties (e.g., the nuclear anapole moment) by more than **12 orders of magnitude** [5]

$$W_{PV} \sim \frac{< - |H_{PV}| + >}{E_- - E_+}$$

Atom: ~1 eV  
Molecule: ~10 μeV  
Molecule + B: ~1 peV

**Factor > 10<sup>12</sup> amplification!**

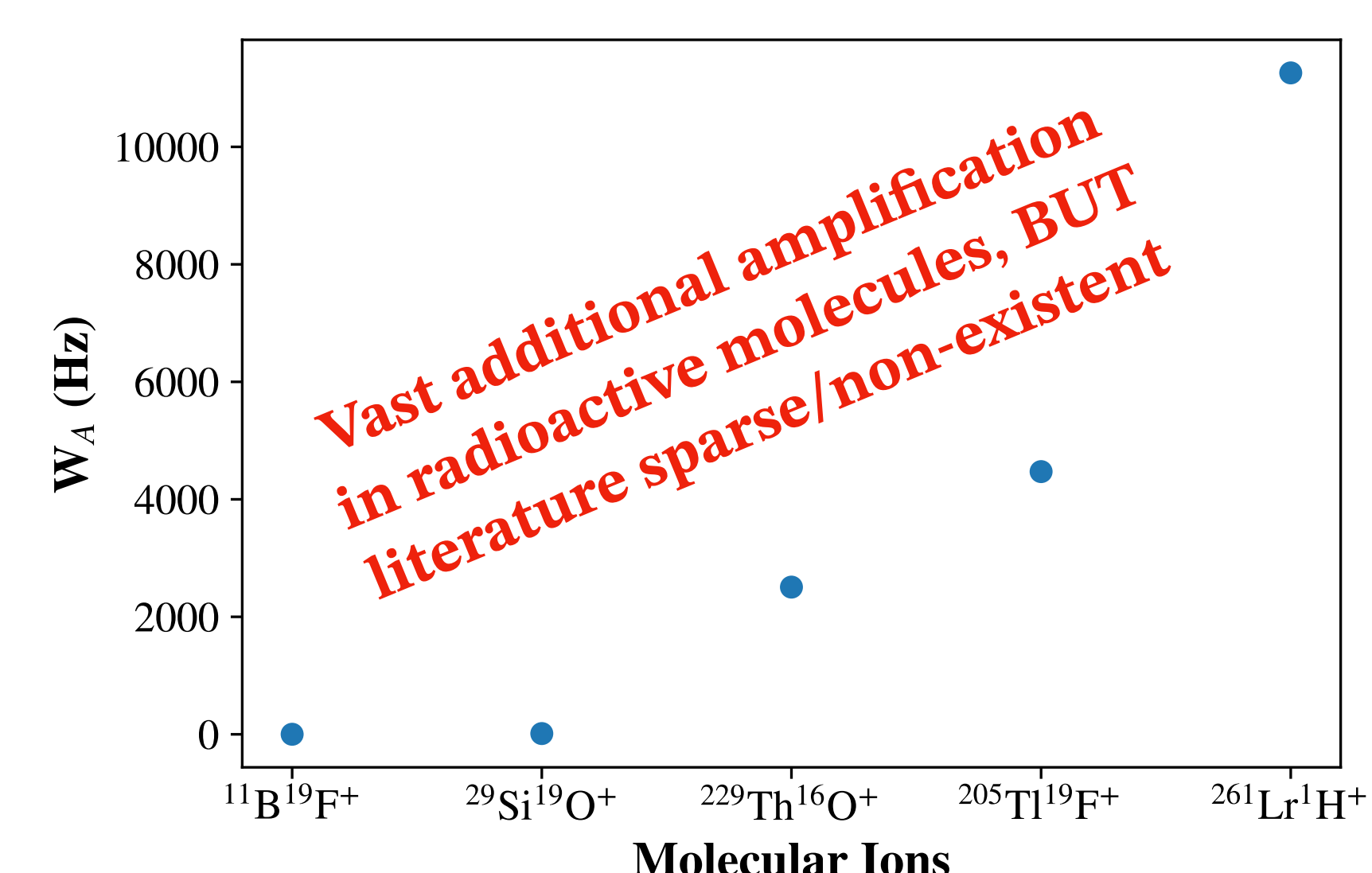


Fig 7. Diatomic molecular ions with sizable weak matrix elements  $W_A \sim \langle - |H_{PV}| + \rangle$  [6]

## References

- [1] Ringle R. Final report: DE-SC0021423 (2024).
- [2] Takamine A et al. *Rev. Sci. Instrument.*, **76**, 103503 (2005).
- [3] Greiner F et al. *NIM-B*, **463**, 324-326 (2020).
- [4] Jacobs A et al. *Int. J. Mass Spec.*, **482**, 116931 (2022).
- [5] Karthein J et al. *Phys. Rev. Lett.*, **133**, 033003 (2024).
- [6] Arrowsmith et al, *Rep. Prog. Phys.*, **87**, 084301 (2024).

## Acknowledgment



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